

Applications of Artificial Intelligence

Unit 2 Consolidate – Individual Task

Chosen case for analysis:

ELIZA (Weizenbaum): --- How does the ELIZA effect manifest in modern conversational agents like ChatGPT or Siri or Alexa?

Analysis

Background: Eliza was an artificial intelligence system that stimulated natural language conversation with a computer.

- ELIZA noticed keywords in the users input through pattern matching and used this to form a response in a pre-fixed manner.

For example:

USER: "I am unhappy"

ELIZA: "Why are you unhappy".

ELIZA was programmed to pick up on the word(s) following "I am" and append them to "Why are you"

Or

USER: "I think that {insert assertion}"

ELIZA: "why do you think that {insert assertion}?"

ELIZA relied heavily on the use of these stock-phrases to prompt further conversation and give an illusion of understanding. (Weizebaum, pg. 37). However, this was enough to make users believe it understood them and develop an emotional bond.

ELIZA lacked a human-like "belief structure". Without an organised network of beliefs, it relied on pattern-matching, text prediction and transformation rules, with no semantic understanding of the users' beliefs.

In modern chat agents, the main similarity is their ability to pick up on keywords. LLMs, however, have wider capabilities.

For example:

User: Alexa, search of {insert text} would prompt Alexa to search for the text after search for.

Or

USER: Siri, play {insert text} would prompt Siri to search for and play insert text or the nearest thing to it available on the music and video playing software.

References:

London Intercultural Academy. (n.d.). *The story of ELIZA: The AI that fooled the world*.
<https://liacademy.co.uk/the-story-of-eliza-the-ai-that-fooled-the-world/> (accessed 16 February 2026)

Weizenbaum, J. (1966). ELIZA—A computer program for the study of natural language communication between man and machine. *Communications of the ACM*, 9(1), 36–45.
<https://doi.org/10.1145/365153.365168>